

Prospective source model of the 2026 M_w 7.4 San José del Palmar (Colombia) intermediate-depth earthquake

A blind test of the MTUQ+WISP pipeline, frozen before the USGS finite fault — and the discovery of a 35-s emergent nucleation phase

Automated pipeline (17.Venezuela_2026)

Frozen 2026-08-11 — no USGS finite-fault product existed at freeze time

Purpose and headline

The M_w 7.4 earthquake of 2026-08-10 beneath western Colombia (110 km deep) has, at freeze time, no published finite-fault model. We freeze a teleseismic-only source model for later head-to-head comparison. Two findings lead: (1) the event is not a Bucaramanga-nest earthquake — it belongs to the Cauca intermediate-depth cluster of the subducting Nazca slab, 407 km southwest of the nest; (2) the rupture began with a ~ 35 -s weak emergent nucleation phase before the main moment release 29 km south of the epicentre — a property that breaks naive origin-anchored inversions and had to be handled explicitly.

1 Event, tectonic setting, and the Bucaramanga question

Origin	2026-08-10T12:34:28.1 UTC, 4.844°N, 76.242°W, depth 110.3 km
Magnitude	M_w 7.4 (W-phase); $M_0 = 1.66 \times 10^{20}$ N m; centroid 120.5 km; DC 87 %
W-phase NP1	124.0/58.6/158.5
W-phase NP2	225.6/71.8/33.2 (NE–SW, steep, oblique reverse)
W-phase centroid	4.585°N, 76.161°W (29 km S of epicentre), 12:35:02.6 = origin + 34.5 s

Bucaramanga nest? No. The nest — the world’s densest intermediate-depth seismic nest — sits at $\sim 6.82^\circ\text{N}$, 73.15°W , 140–160 km depth beneath Santander, NE Colombia. This event is 407 km to the southwest at 4.84°N , 76.24°W , 110–120 km deep: the Cauca intermediate-depth cluster of the subducting Nazca slab (cf. Chang et al. 2019), a different slab segment with its own seismicity. It ranks among the largest instrumental events of that segment.

2 Data

120 three-component stations (316 channels), 30–90°, maximum azimuthal gap 47° (the Pacific). MTUQ used an azimuth-balanced 36-station response-corrected subset with the deep-event bands (body 25–60 s, mantle surface waves 200–450 s). Both nodal planes extend below the 126.5-km

WISP surface-wave GF cap, so the finite-fault plane comparison is body-only on both planes — fair by construction (the San Pedro de Atacama rule).

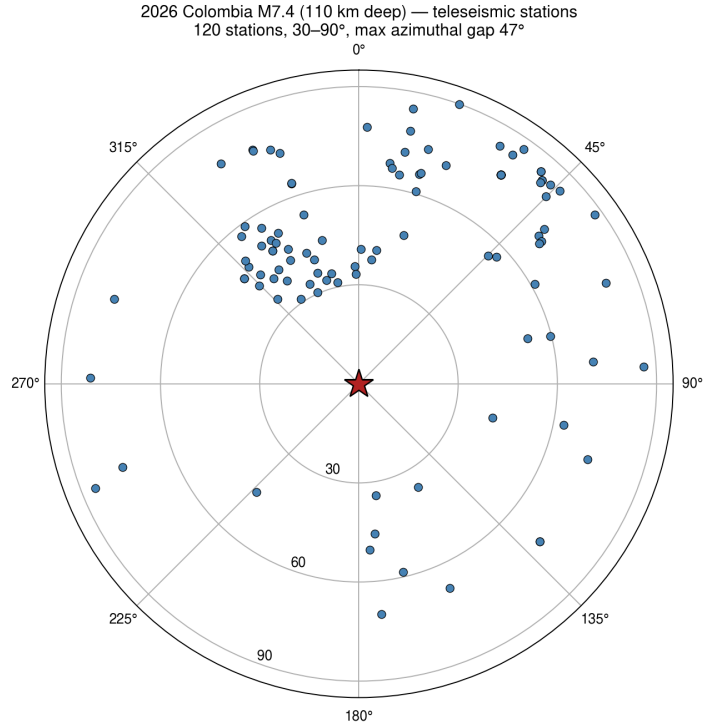


Figure 1: Teleseismic stations (azimuthal-equidistant; rings 30/60/90°).

3 The emergent nucleation phase — and why origin-anchored inversions fail

The observed P wavetrains begin weakly and emergently; the main impulsive packet arrives only $\sim 45\text{--}55$ s after origin (its pP/sP depth-phase echoes at $\sim 75\text{--}90$ s). Independently, the W-phase centroid time lags the origin by 34.5 s and sits 29 km south. The implied nucleation-to-asperity apparent velocity is $\sim 0.8 \text{ km s}^{-1}$ — far below any physical rupture speed — so a finite fault nucleating at the cataloged hypocentre at the cataloged time cannot place the moment where the data demand it. Our first (origin-anchored) inversion demonstrated the failure exactly: slip piled at the fault edge (86% of peak in an edge row — a violated containment rule), the STF peaked at an unreachable 40 s, and the synthetics fit only the long-period envelope. Origin-cut MTUQ windows likewise missed the main energy (M_w pinned at the grid edge).

The fix (the operational Yeck-style relocation): re-anchor the entire analysis — window cutting, Green’s functions, fault geometry, nucleation — at the W-phase centroid in space and time. The 35-s precursor then lies outside the modelling window (its moment is a few per cent at most), and the main rupture is imaged about its true origin.

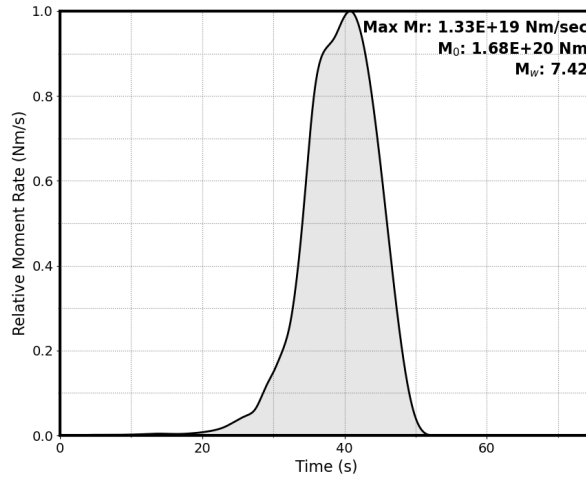


Figure 2: The INVALID origin-anchored model’s STF, shown as the diagnostic: the peak at ~ 40 s is kinematically unreachable from the hypocentre, so the inversion smeared slip to the fault edges.

4 Finite fault (WISP, centroid-anchored): both planes, **shift_match**, body-only

plane (W-phase labels)	misfit (post-shift)	edge slip top/bottom	peak slip
226/72/33 (preferred)	0.1363	1 % / 3 %	1.63 m
124/59/159 conjugate	0.1485	40 % / 2 %	1.12 m

Table 1: An 8.2 % margin plus the conjugate’s top-edge violation = a near-decisive plane call for the NE–SW steep oblique-reverse plane — exactly where the rupture-size criterion (Calama 1.6 % tie at 10 km; SPdA 10 % decisive at 50 km) predicts discrimination turns on. WISP’s directory labels are swapped vs USGS (WISP NP1 = W-phase NP2), as at Kumamoto. Recovered $M_0 = 1.68 \times 10^{20}$ N m (W-phase 1.66).

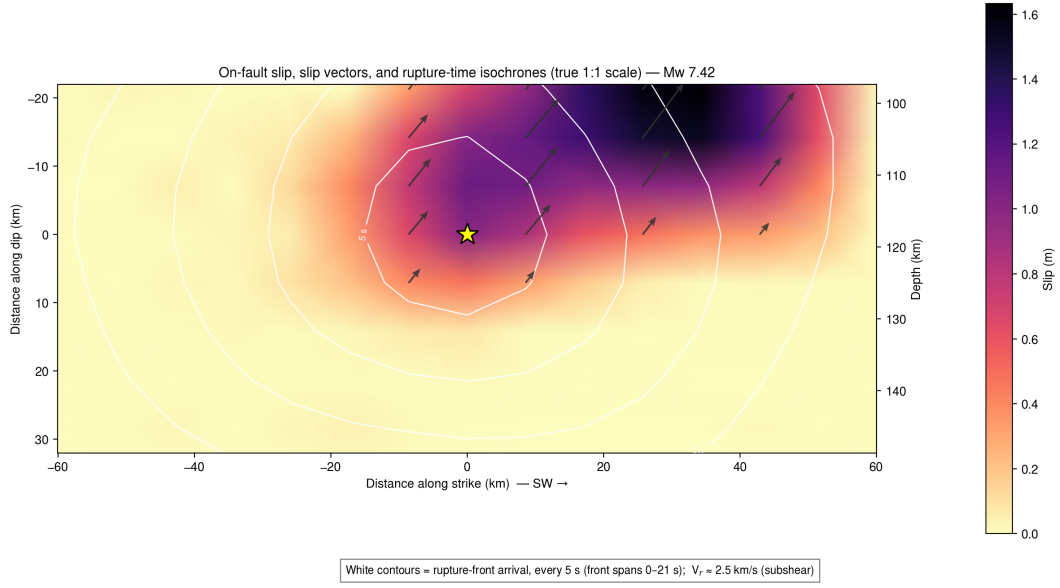


Figure 3: Preferred plane (226/72/33), true scale: a single dominant asperity ~ 17 km SSW of the centroid nucleation (~ 45 km SSW of the epicentre), up-dip, slip at 85–125 km depth, peak 1.63 m, cleanly contained (edge slip 1–3%). Effective area $2695 \text{ km}^2 \rightarrow$ radius 29 km $\rightarrow$ stress drop ≈ 2.9 MPa.

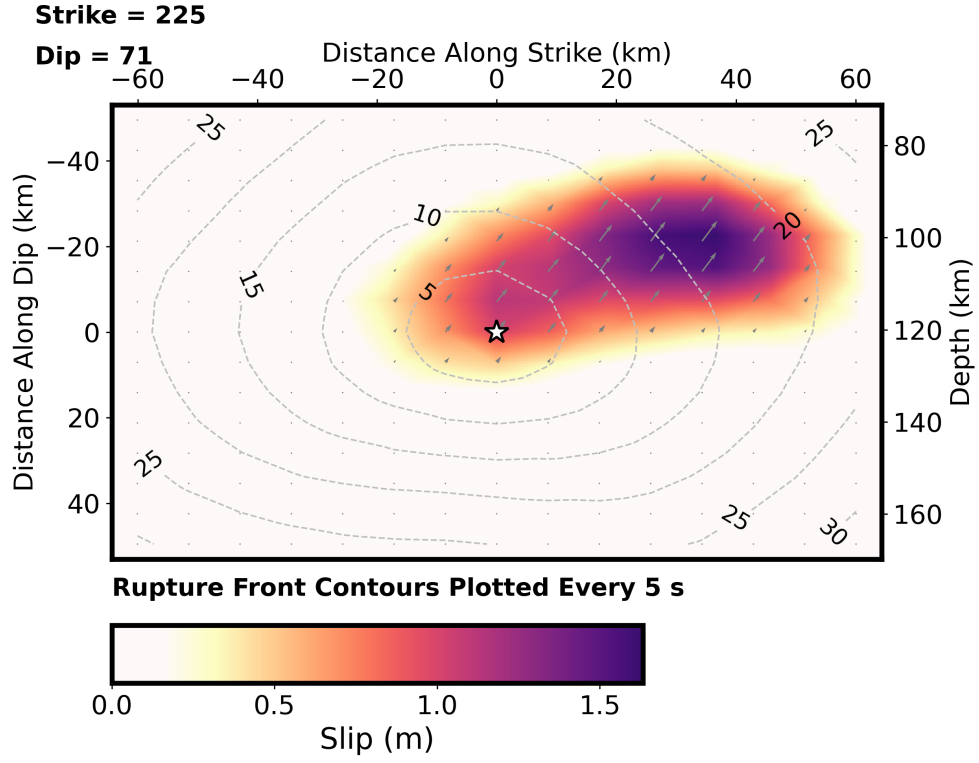


Figure 4: Same model with 5-s rupture-front isochrones: main-phase duration $T_{5-95} \approx 18.7$ s (times relative to the centroid nucleation; the ~ 35 -s precursor precedes this and is below our band).

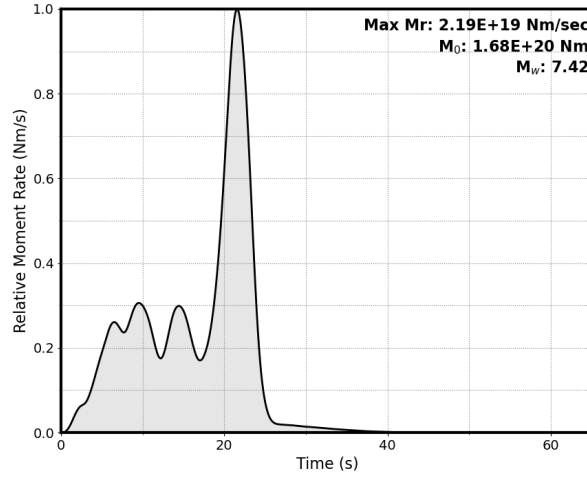


Figure 5: Main-phase moment-rate function (centroid-anchored, preferred plane).

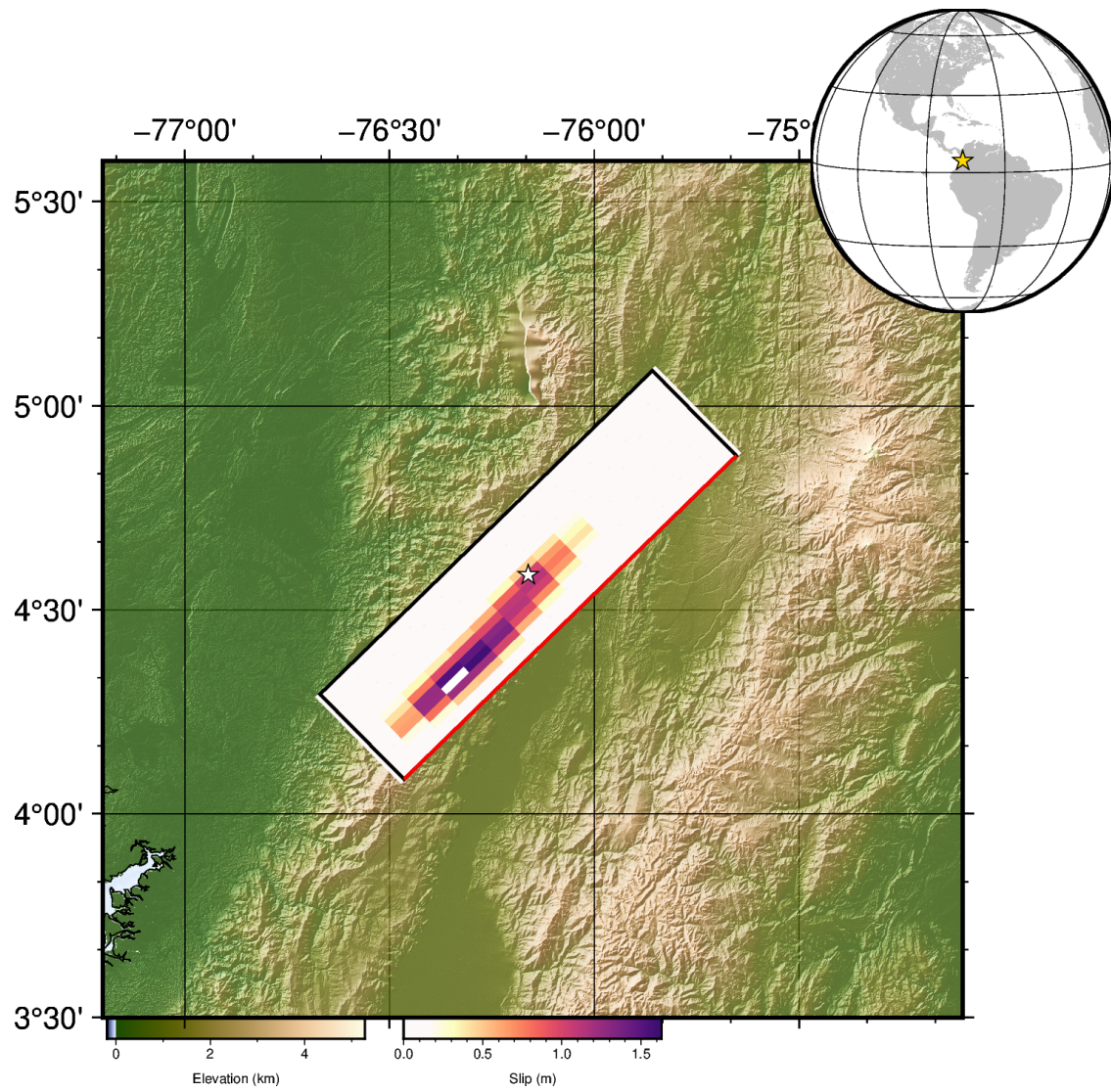


Figure 6: Map view of the preferred-plane slip.

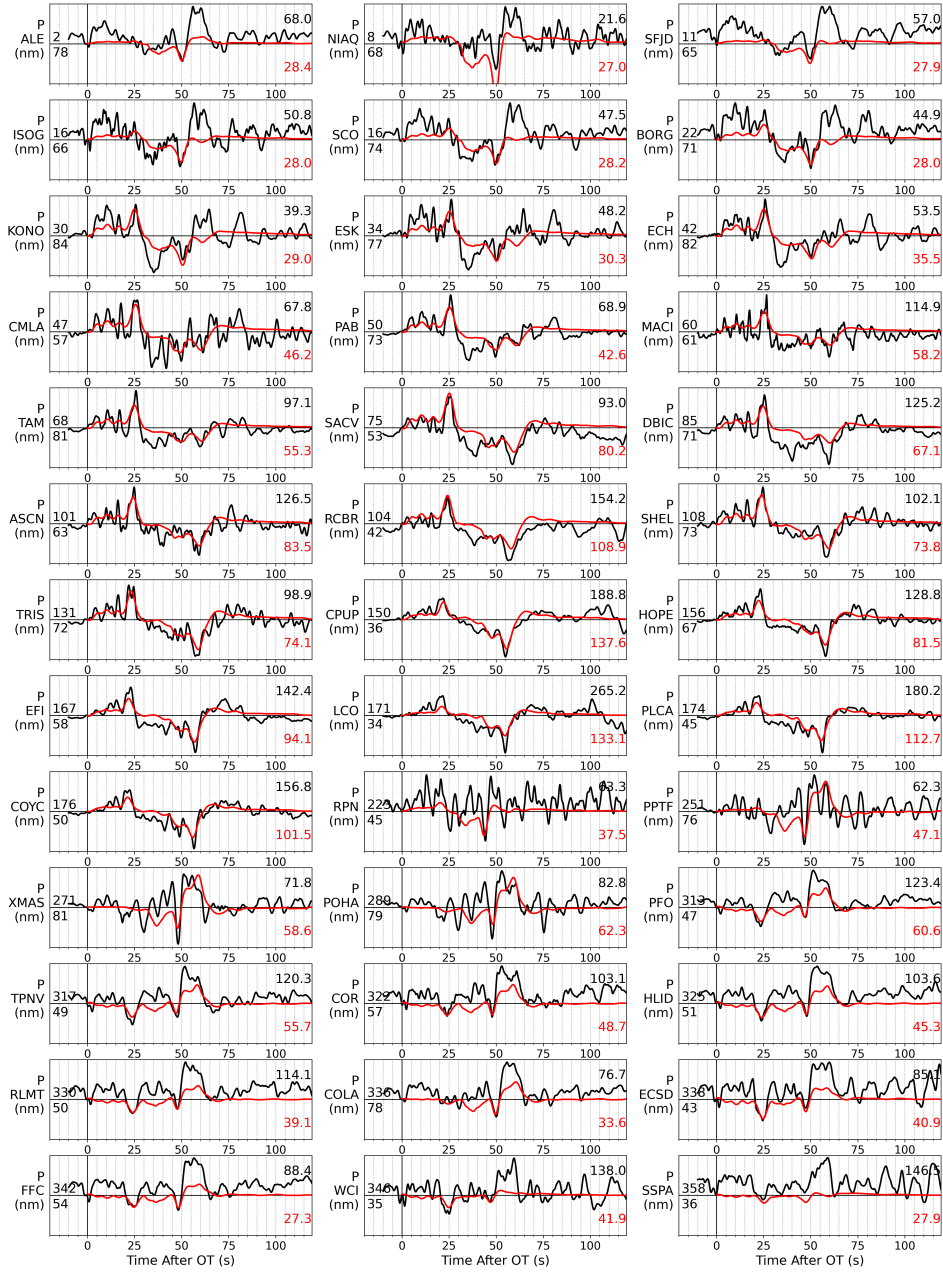


Figure 7: Teleseismic P fits (preferred plane, shifted): the main packet and its depth phases are now aligned and fit.

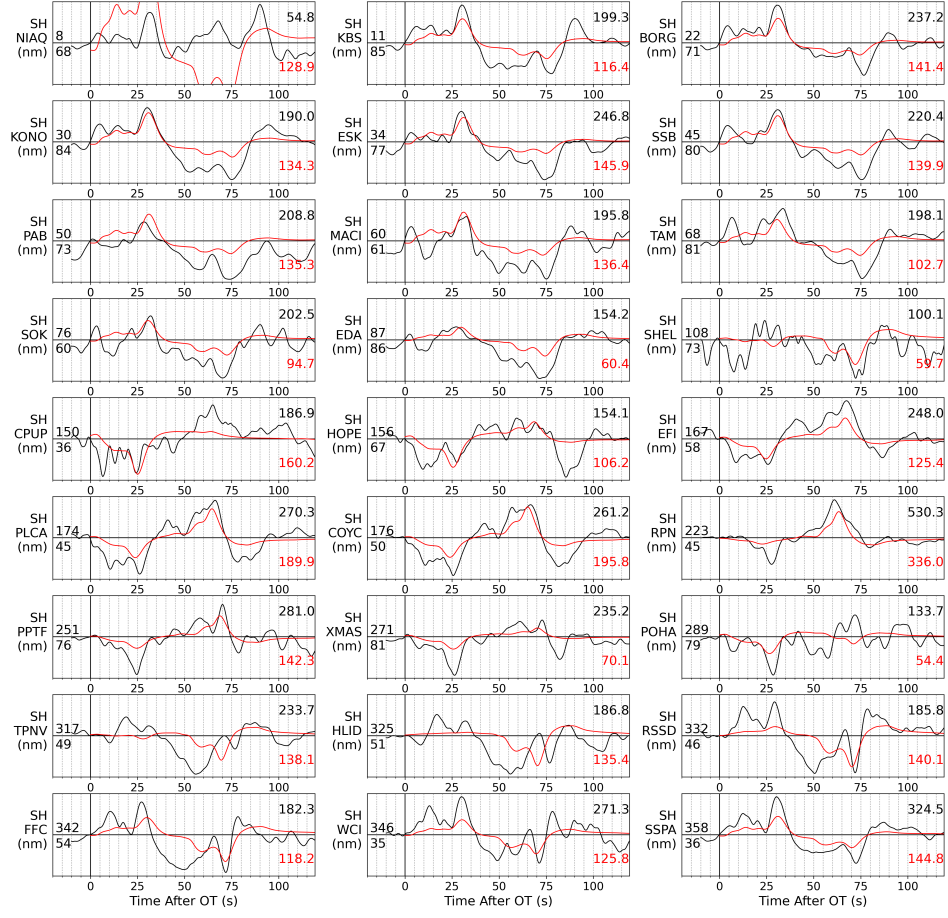


Figure 8: Teleseismic SH fits (preferred plane, shifted).

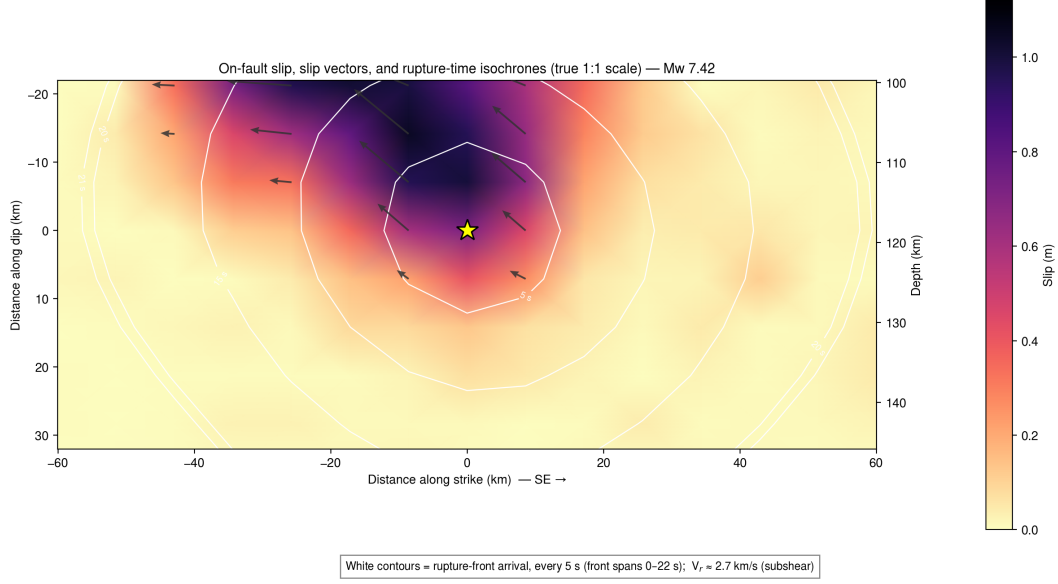


Figure 9: Conjugate plane (124/59/159) for comparison: 8.2 % worse fit and 40 % of peak slip at its top edge.

5 Moment tensor (MTUQ): the point source breaks on this complex source

variant	mechanism	M_w	Kagan vs W-phase
joint DC @120.5 km	202/81/−65	7.10 (grid edge)	95.3°
body P+SH @120.5 km	220/61/−52	7.10 (grid edge)	83.6°
body depth scan (60–150 km)	minimum at 70–90 km	7.05–7.25	unstable 28.6–93°

Table 2: The body-wave depth curve bottoms at 70–90 km — shallower than the W-phase centroid — and the mechanism is stable nowhere except ~ 90 km, where it snaps to 225/71/4 (28.6°). A 40-km-offset late asperity, a 35-s precursor and a 40-km-tall slip zone violate the point-source approximation. The robust element — strike ~ 207 – 225° — matches the preferred finite-fault plane. Pipeline rule: for complex sources, take mechanism and depth from the W-phase and spend the teleseismic data on the finite fault.

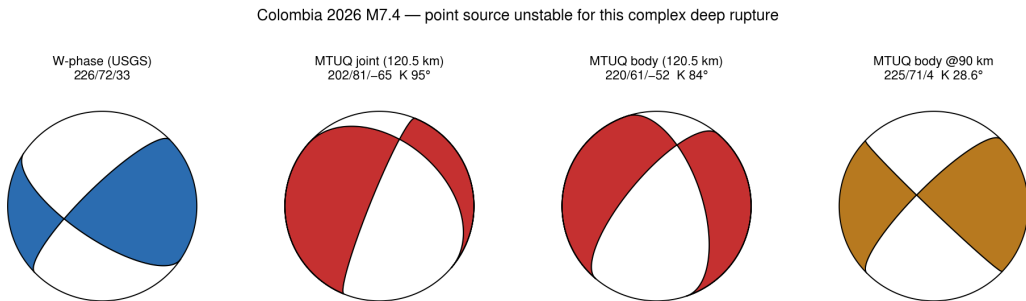


Figure 10: W-phase reference vs MTUQ attempts.

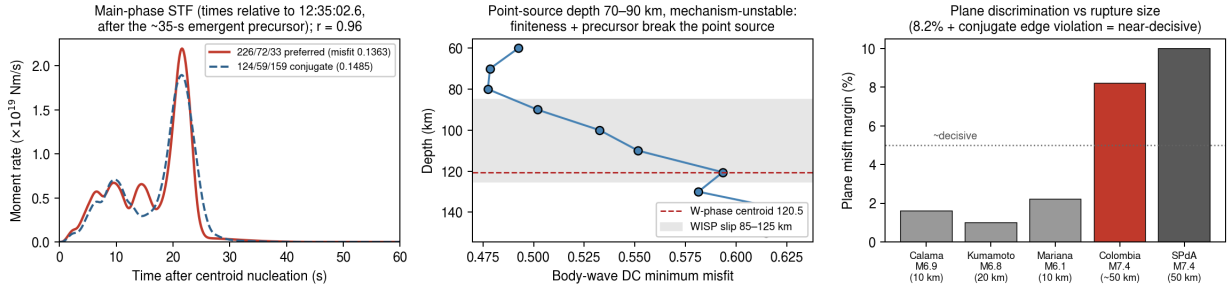


Figure 11: Left: the main-phase STF is stable across the two assumed planes. Centre: the point-source depth curve vs the finite-source depth extent — the point source is biased shallow by finiteness. Right: the plane-discrimination margin in the pipeline’s rupture-size context.

6 Directivity: two-stage, unilateral southward + up-dip

(A correction made within the same pre-USGS freeze window: the first push described the asperity as “NE along strike” — an axis-misread of the WISP plot; the geographic computation is authoritative.)

1. Model-independent stage 1: epicentre $\rightarrow$ W-phase centroid = 30 km toward azimuth 163° (SSE), 110 $\rightarrow$ 120.5 km, in ~ 34.5 s — apparent 0.9 km s^{-1} (the emergent nucleation phase).
2. Preferred plane stage 2: the early $\rightarrow$ late moment centroid moves 27 km toward azimuth 220° (SW) while shallowing 110 $\rightarrow$ 101 km; 75 % of the moment lies forward of the nucleation vs 24 % behind — predominantly unilateral, along-strike SW plus up-dip.
3. Robustness: the up-dip component appears on both planes; the conjugate’s horizontal direction (northward) contradicts the model-independent stage-1 offset, further disfavouring that already-rejected plane.

7 Frozen prospective prediction

- Fault plane: 226/72/33 (NE–SW, steep, oblique reverse) — 8.2 % margin plus the conjugate’s edge violation.
- Two-stage rupture: a ~ 35 -s weak emergent nucleation from the cataloged hypocentre, then the main asperity ~ 45 km SSW of the epicentre, up-dip, 85–125 km depth; peak slip ≈ 1.6 m; main-phase $T_{5-95} \approx 19$ s; $\Delta\sigma \approx 2.9$ MPa.
- M_0 1.68×10^{20} N m (M_w 7.42), matching the W-phase to 1 %.
- Scorecard when USGS posts: plane choice; whether they relocate the hypocentre/nucleation; asperity position (SSW + up-dip); depth extent; peak slip; duration.

Caveats. Teleseismic-only; the precursor's own moment and location are below our band and not imaged; the mechanism statement rests on the W-phase (MTUQ instability documented above); body-only fits on both planes (surface-wave GF cap).